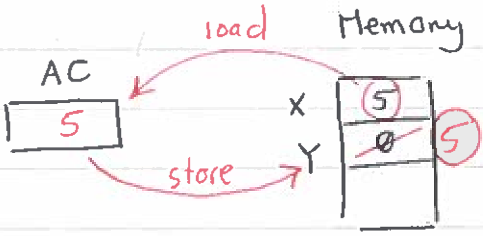
* Simple MARIE programs

① write the following code segment in MARIE assembly lang.

X = 5 ;
Y = X ;

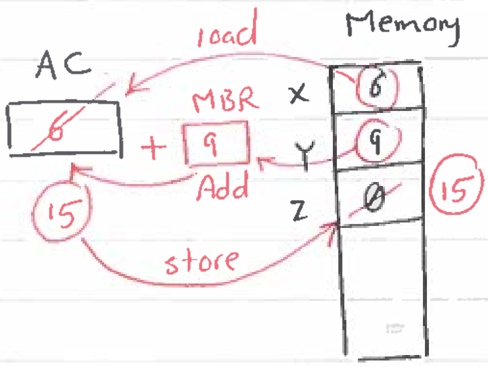
```
Load X
Store Y
Output  →  5
Halt
X, DEC 5
Y, DEC 0
```



② write the following code segment in MARIE assembly code

X = 6
Y = 9
Z = X + Y

```
Load X
Add Y
store Z
Output  →  15
Halt
X, DEC 6
Y, DEC 9
Z, DEC 0
```



③ write the following code segment in MARIE assembly code

X = 5
Y = 9
Z = X − Y

```
load X
subt Y
store Z
Output  →  -4
Halt
X, DEC 5
Y, DEC 9
Z, DEC 0
```

RTN

0001
① Load X
MAR ← X
MBR ← M[MAR]
AC ← MBR

0010
② Store X
MAR ← X, MBR ← AC
M[MAR] ← MBR

0011
③ Add X
MAR ← X
MBR ← M[MAR]
AC ← AC + MAR

0100
④ Subt X
MAR ← X
MBR ← M[MAR]
AC ← AC − MAR

⑤ Input 0101
⑥ output 0110
⑦ Halt 0111
⑧ skipcond 1000
⑨ Jump 1001

* Fetch - Decode - Execute Cycle

The steps that a computer follows to run a program

[1] CPU fetches an instruction (from MM to IR)
[2] Decodes it (determine opcode)
[3] Executes it (perform the operation(s))

In Details :-

① Copy the content of PC to MAR (MAR ← PC)

② Fetch instruction from Memory found in MAR, placing the instruction in IR, increment PC by 1. PC now points to the next address

$$IR \leftarrow M[MAR]$$
$$PC \leftarrow PC+1$$

③ Copy the 12 bits address of IR into MAR, decode the 4 bits to determine the opcode.

$$MAR \leftarrow IR[11-0]$$
$$Decode \leftarrow IR[15-12]$$

④ Execute the instruction

$$MBR \leftarrow M[MAR]$$
$$AC \leftarrow MBR$$

Arab Academy for Science, Technology & Maritime Transport
College of Computing and Information Technology

- **Course Title : Introduction to Computer Architecture**
- **Chapter 04 : MARIE An Introduction to a Simple Computer**
- **Course Code : CE243**
  Section 05

Example:

Trace the following program to Add two numbers (Book page 250, 251)

Hex Address	Instruction	Hex Content of Memory
100	LOAD 104 X	1104
101	ADD 105 Y	3105
102	STORE 106 Z	2106
103	Halt	7000
104	X 0023	0023
105	Y 00F5	00F5
106	Z 0000	0118

1. LOAD 104

Step	RTN			PC	IR	MAR	MBR	AC
Initial Values				100	-	-	-	-
Fetch	MAR	←	PC	100	-	100	-	-
	IR	←	M[MAR]	100	1104	100	-	-
	PC	←	PC+1	101	1104	100	-	-
Decode	MAR	←	IR[11-0]	101	1104	104	-	-
	Decode IR [15-12]			101	1104	104	-	-
Get Operand	MBR	←	M[MAR]	101	1104	104	0023	-
Execute	AC	←	MBR	101	1104	104	0023	0023

2. Add 105

Step	RTN			PC	IR	MAR	MBR	AC
Initial Values				101	1104	104	0023	0023
Fetch	MAR	←	PC	101	1104	101	0023	0023
	IR	←	M[MAR]	101	3105	101	0023	0023
	PC	←	PC+1	102	3105	101	0023	0023
Decode	MAR	←	IR[11-0]	102	3105	105	0023	0023
	Decode IR [15-12]			102	3105	105	0023	0023
Get Operand	MBR	←	M[MAR]	102	3105	105	00F5	0023
Execute	AC	←	AC + MBR	102	3105	105	00F5	0118

3. Store 106

Step	RTN			PC	IR	MAR	MBR	AC
Initial Values				102	3105	105	00F5	0118
Fetch	MAR	←	PC	102	3105	102	00F5	0118
	IR	←	M[MAR]	102	2106	102	00F5	0118
	PC	←	PC+1	103	2106	102	00F5	0118
Decode	MAR	←	IR[11-0]	103	2106	106	00F5	0118
	Decode IR [15-12]			103	2106	106	00F5	0118
Get Operand	-			103	2106	106	00F5	0118
Execute	MBR	←	AC	103	2106	106	0118	0118
	M[MAR]	←	MBR	103	2106	106	0118	0118